



"THE FUTURE OF LIFE DEPENDS TO A LARGE EXTENT ON CREATING NEW AND SYNTHETIC LIFE FORMS."

J. Craig Venter

based on the chromosome we had put in. This process, I believe, is going to be the enabling step to take us to the next stage. It will help us understand evolution in a way that has never been studied before.

We are now trying to get it to work with chromosomes we've made in the laboratory, to see if we can

boot up a cell with it. Along the research route we saw that apart from answering fundamental questions of science, this could provide additional benefits to society. Biology could be utilised to address every area of our lives—from medical treatment to renewable sources of fuels. You could make large amounts of biofuels, derived synthetically by bacteria, instead of taking carbon from the ground and burning it—something that is the biggest threat to humanity. In other words, dramatic climate changes due to continued replacement of carbon in our atmosphere from burning fossil fuels.

What is the value of these discoveries? A better understanding of life and its evolution on earth. The new field of synthetic biology can change the way we think about life. We can use millions of newly-discovered organisms and genes to tell us how the environment is changing. And it can show that it is possible to use living systems to increase our chances of survival as a species.

D I S C U S S I O N

Q. You have been called a controversial biologist. What's your take on that?

Venter: If you are not at the leading edge of your field, you can't progress. And if you are making real progress it will be controversial. So the more I see that label, the more proud I feel.

Q. How can you be sure that your diploid sequence is actually accurate and further research won't show up something else?

Venter: As yet no human genome sequence is truly complete. But the exciting thing about some of the new technologies is that they are building on the reference sequence that we published. The accuracy can only get higher and higher over time.

Q. You talked about capturing phenotypical information. But even clinically those are difficult

to diagnose. How do you hope to do that genetically?

Venter: That's the biggest challenge of modern genetics. We need well-characterised phenotypic bits of information that can be digitised. We are talking to groups around the world—specialists in different areas—to help put together the criteria to set a world standard, for what should be done to collect such information. That's the challenge of the next 10 years.

Q. How long will it take for us to create better children, in terms of intelligence and health?

Venter: I hope we will wait until we have sufficient knowledge to do so intelligently, if that is possible. Right now, we can't even characterise well the phenotypes of individuals. And I have not met a human yet, who I think should be cloned. So trying to define characteristics with

which we want to alter the race genetically, won't happen and shouldn't happen in this century. Also that would mean losing out on people with *The Beautiful Mind*.

Q. On the energy potential of your work, isn't there a risk of complacency? Since fossil fuels will be replaced, people may stop saving energy. When and how will we meet our energy needs?

Venter: You mean, we shouldn't talk about breakthroughs if they don't lead to change in people's behaviour? I have always advocated the need for every type of breakthroughs in every field of science and engineering. Biology will provide part of the solution. Some of the ideas are years, if not decades away. But the main challenge is still in the scale. It's hard to appreciate the scale at which we burn carbon. So one of the notions that we are trying to

develop is what Edison advocated: generate and use power locally instead of transmitting it long distance.

Q. Can synthetic biology have a negative impact on the ecology, by producing harmful wastes?

Venter: No one is talking about releasing organisms into the environment. We now know enough about biology to be able to design organisms that can't live outside a defined chemical system.

Q. Did you find anything in your own gene sequence that makes you as brilliant as you are?

Venter: I wish I could say yes. As I wrote in my autobiography, *Life Decoded*, mostly environmental interactions shaped my growth. I was lucky to have started with pretty good parents, though.

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